

TECHNOLOGY STACK

Project Name: ComicCraft – AI Powered Comic Story Generator

Team ID: _____

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Jinja2 Templates	Provides a responsive and user-friendly interface for comic generation, preview, and download. HTML, CSS, and JavaScript ensure compatibility across browsers, while Jinja2 enables dynamic rendering of generated comic content.
2	Backend / Server-Side	Python, FastAPI	FastAPI offers high performance, easy API development, asynchronous processing, and seamless integration with AI models. Python provides extensive support for Generative AI and image-processing libraries.
3	Database / Data Storage	SQLite (Development), PostgreSQL (Production)	SQLite enables lightweight local development, while PostgreSQL offers scalable and reliable storage for user data, generated comics, and application metadata.
4	Cloud / Hosting / Deployment	Docker, Render / AWS EC2	Docker ensures consistent deployment across environments. Render or AWS EC2 provides scalable cloud hosting and supports future growth of the platform.
5	Version Control & CI/CD	Git, GitHub, GitHub Actions	Enables collaborative development, source code management, version tracking, automated testing, and deployment workflows.
6	Third-Party APIs / Other Tools	Google Gemini API, Hugging Face Inference API, Pillow, FPDF	Gemini API generates comic stories, dialogues, and narration. Hugging Face APIs generate comic illustrations. Pillow handles image processing, while FPDF creates downloadable comic PDFs.

Frontend Technologies

HTML5

Used to structure web pages and user interfaces.

CSS3

Provides styling, responsiveness, and visual appeal.

JavaScript

Handles user interactions and dynamic content updates.

Jinja2

Renders dynamic comic content generated by the backend.

Backend Technologies

Python

Primary programming language used for AI integration and backend development.

FastAPI

Used to create REST APIs for comic generation and PDF export with high performance and scalability.

Database Technologies

SQLite

Used during development for simplicity and rapid prototyping.

PostgreSQL

Used in production environments for reliable and scalable data storage.

AI & Generative Technologies

Google Gemini API

Responsibilities:

- Story generation

- Dialogue generation
- Narration generation
- Comic panel descriptions
- Prompt enhancement

Hugging Face Inference API

Responsibilities:

- AI image generation
 - Character illustrations
 - Comic scene generation
 - Art style customization
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PDF & Image Processing Tools

Pillow

Used for:

- Image resizing
- Image manipulation
- Comic panel formatting

FPDF

Used for:

- PDF generation
 - Comic book export
 - Multi-page comic creation
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Deployment Technologies

Docker

Benefits:

- Environment consistency
- Easy deployment
- Scalability

Render / AWS

Benefits:

- Cloud hosting
 - High availability
 - Future scalability
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Version Control & Collaboration

Git

Used for source code version management.

GitHub

Used for:

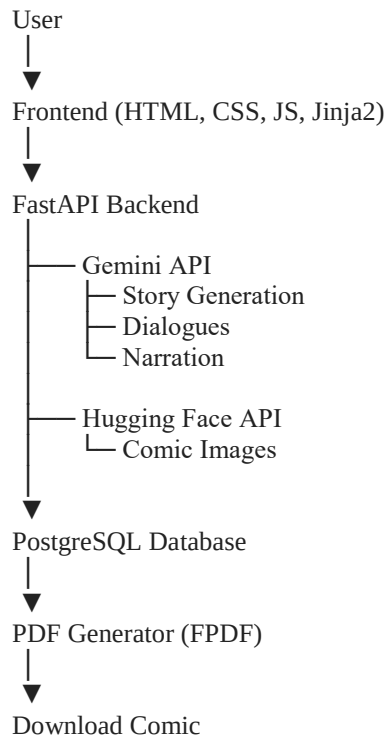
- Code repository hosting
- Team collaboration
- Issue tracking
- Pull requests

GitHub Actions

Used for:

- Continuous Integration (CI)
 - Continuous Deployment (CD)
 - Automated testing
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Technology Architecture Overview



Conclusion

The selected technology stack for ComicCraft combines modern web technologies, cloud-ready deployment tools, scalable backend frameworks, and powerful Generative AI services. This stack ensures that the application remains efficient, maintainable, secure, and capable of generating high-quality AI-powered comics while supporting future enhancements and large-scale deployment.